

```

unsigned int* up;    /*4B*/
short x;            /*2B*/
char pad2[2];
int v;              /*4B*/
char k;             /*1B*/
char pad3[3];
};

```

Now our structure looks like what's shown below:

BYTE1	BYTE2	BYTE3	BYTE4
C	pad1	pad1	pad1
Up	up	up	up
x	x	pad2	pad2
v	v	v	v
k	pad3	pad3	pad3

The best way to decrease the slope is by 'reordering the structure' members in decreasing lengths of variables. If we reorder the structure, we get the following:

```

struct ESTR
{
    unsigned int* up;    /*4B*/
    int v;              /*4B*/
    short x;            /*2B*/
    char c;             /*1B*/
    char k;             /*1B*/
};

```

Now, our structure looks like what's shown below:

BYTE1	BYTE2	BYTE3	BYTE4
Up	up	Up	up
v	v	v	v
x	x	c	k

You can see the structure has come down to 12B from 20B. But do not think that padding can be completely eliminated. In this specific example, it has happened with reordering. This may not happen always. But the right practice is to sort the members according to their size.

The disadvantage of reordering

Reordering may affect cache-locality. For example, in a structure, you refer to two variables very frequently. But because of reordering, they are placed very far in terms of word number. So when they are mapped in cache, they become part of different blocks which generate cache-miss and require very frequent cache block swapping. Hence, execution becomes slow.

'Structure packing' stops compilers from adding any pad but forces the use of a defined order. It can be achieved with '#pragma pack' in C. Then our first structure alignment looks like what follows:

BYTE1	BYTE2	BYTE3	BYTE4
c	up	up	up
up	x	x	v
v	v	v	k

This is called structure packing.

Now, if the CPU wants variable v or up it needs two word accesses for each word consisting of 4 bytes.

So it is the design, which will enable users to make a choice. In embedded systems, there is a performance versus memory size trade-off, because everybody expects faster responses but the size of the memory is fixed. Fragmentation may also not be handled in an embedded OS because of the size constraints. So, the designer has to be careful. **END** 

By: Supriyo Ganguly

The author is an engineer working at the Electronics Corporation of India Limited, Hyderabad.

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